Learning with Local and Global Consistency: Group 28 Report (draft)

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1 Introduction

[D to write: the semi-supervised setting, local vs global consistency assumptions, why the two-moons picture motivates the method. Target ≈0.4 page. Placeholder keeps the page budget visible in the draft.] ***[TODO: placeholder]***

2 Algorithm

[A to write: the four-step construction, spreading-activation reading, self-reinforcement note, the update rule that §3 then analyses. Target ≈0.5 page. Cross-reference §3 for why it converges.] ***[TODO: placeholder]***

3 Theory: convergence and regularization

The iteration of Section 2, $F(t{+}1) = \alpha S F(t) + (1-\alpha)Y$ with $S = D^{-1/2} W D^{-1/2}$, converges because of what the symmetric normalization buys: S is similar to the row-stochastic matrix $D^{-1}W$, so its eigenvalues share those of a stochastic matrix, and being symmetric they are real, all in $[-1, 1]$ with top eigenvalue exactly 1. Hence for $\alpha \in (0, 1)$ the spectral radius of αS is $\alpha < 1$, the Neumann series $\sum_{i \geq 0} (\alpha S)^i = (I - \alpha S)^{-1}$ converges, and iterating from any starting point gives

$$F^* = (1 - \alpha)\,(I - \alpha S)^{-1}\, Y, \tag{1}$$

the paper's closed form (its Eq. 3 up to the constant $1-\alpha$, which does not affect the argmax rule). Writing $F(t) - F^* = (\alpha S)^t\,(F(0) - F^*)$ shows both advertised properties at once: the limit is independent of $F(0)$, and the error decays geometrically at rate α.

Equation (1) is not a lucky coincidence of the iteration; it is the unique global minimiser of a regularised objective. With $\mathcal{Q}(F) = \frac{1}{2}\sum_{i,j} W_{ij} \left\| \frac{F_i}{\sqrt{D_{ii}}} - \frac{F_j}{\sqrt{D_{jj}}} \right\|^2 + \mu \sum_i \|F_i - Y_i\|^2$ (the paper's Eq. 4), setting $\partial \mathcal{Q} / \partial F = 0$ gives $F - SF + \mu(F - Y) = 0$, and the substitution $\alpha = 1/(1+\mu)$, $\beta = \mu/(1+\mu) = 1 - \alpha$ turns it into $(I - \alpha S)F^* = \beta Y$: exactly the iteration's fixed point. The first term is the graph-Laplacian smoothness energy in the degree-normalised coordinates (it equals $\mathrm{tr}(F^\top (I - S) F)$), the second keeps predictions near the clamped labels, and μ is the familiar regularisation trade-off strength: ridge-regression structure with the Laplacian in place of the weight penalty. The same normalisation appears one step earlier in spectral clustering [Ng et al., 2001], and the manifold-regularisation framework of Belkin and Niyogi [2004] is the general form this paper instantiates on a graph; compared to the harmonic-function method of Zhu et al. [2003], this algorithm keeps the clamping term instead of imposing harmonicity.

Verification. We checked these claims numerically rather than trusting the derivation (`experiments/convergence_chec` results in `results/convergence_check_summary.md`). Iterating to the residual tolerance 10^{-12} agrees with solving the linear system directly to 8×10^{-11} max-abs on Role C's ten fixed two-moons trials, with zero argmax disagreements; the observed stopping error equals $\mathrm{tol}/(1-\alpha)$ in every run, because the residual criterion $\|F(t+1) - F(t)\|$ carries an extra factor $(1-\alpha)$ relative to the error $\|F(t) - F^*\|$. Stationarity of $\mathcal{Q}$ at F^* holds to 2×10^{-17}, and every random perturbation of F^* tested raised $\mathcal{Q}$. The measured error-decay slope sits slightly steeper than the predicted $\ln \alpha$ (-0.0132 vs -0.0101 at $\alpha = 0.99$): the α^t worst case rides on the degree-weighted top eigenvector $D^{1/2}\mathbf{1}$, whose component of $F(0) - F^*$ nearly cancels here, so the next eigenvalue dominates in practice (Figure 1).

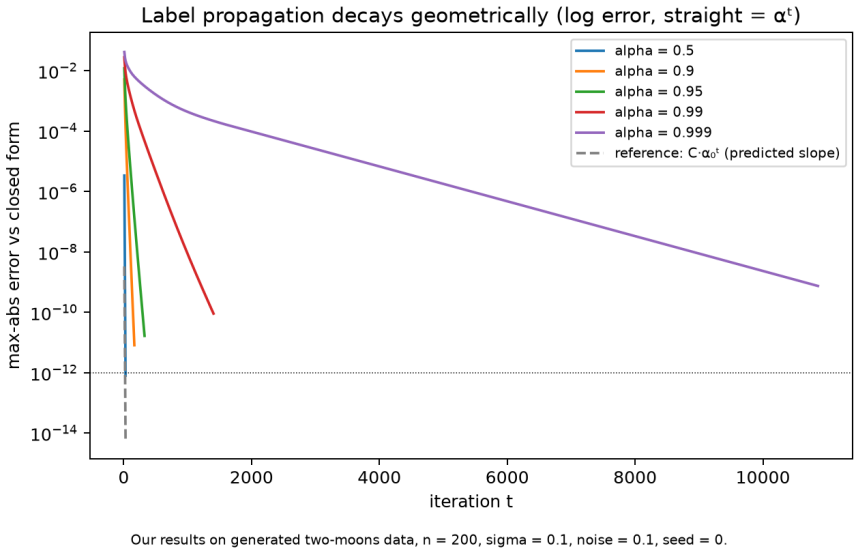


Figure 1: Log max-abs error of the iteration against the closed form, for four values of α, on our generated two-moons data. Straight lines confirm the geometric decay; the dashed line is the predicted α_0^t rate. Our result, not the paper's.

An equation check. The last paragraph of Section 2 of the paper claims the transpose variant is "equivalent" to $F^* = (D - \alpha W)^{-1} DY$. As written it is not, as matrices: measured relative gap 0.82 on our graph. The identity that does hold (verified to 3×10^{-15}) is $(I - \alpha P^\top)^{-1} Y = D\,(D - \alpha W)^{-1} Y$ with $P = D^{-1}W$; pre-multiplying by D is a positive per-row rescaling, so *classifications* agree everywhere and the paper's word "equivalent" is defensible for the argmax rule while the displayed formula misplaces the D factor. All operator variants (S-based, P-based, $P^\top$-based, and $(D - \alpha W)^{-1}$ with or without DY) differ by at most 6/200 predicted points on our toy, a useful robustness observation: the method's classification is far less sensitive to the normalisation choice than the raw score matrices are. Full audit in `results/variants_audit.csv`.

3.1 Experimental evidence and its limits

Zhou et al. Zhou et al. [2003] evaluate their method using a two-moons illustration, a four-class subset of USPS handwritten digits (3,874 examples), and four newsgroup topics (3,970 documents represented by 8,014 features). The real-data comparisons include 1-nearest neighbor, one-versus-rest RBF SVMs, a harmonic Gaussian field method with class mass normalization, and two consistency variants. Error is measured on unrevealed labels, with the digit and text results averaged over 100 trials and at least one labeled example per class. The consistency methods use $\alpha = 0.99$.

Figure 5 of the original paper shows substantially lower digit-recognition error for the main consistency method than for the supervised baselines. The text results are more qualified: the harmonic method performs better at some label budgets. These findings support exploiting unlabeled structure under the tested conditions, rather than universal superiority over other semi-supervised methods. The authors' explanation of the harmonic method's nonmonotonic text results emphasizes the reliability of class-proportion estimates used in class mass normalization.

A central limitation is model selection. The authors use each algorithm's respective optimal parameters while acknowledging the difficulty of choosing parameters with very few labels. The reported comparisons therefore leave practical label-scarce tuning unresolved. In addition, ensuring label coverage of every class is a favorable assumption, and the absence of displayed error bars in Figure 5 limits interpretation of small differences between curves. These are our assessments of the evidence, not additional experimental findings.

3.2 Our partial reproduction on two moons

For each of ten fixed seeds (0–9), we generated 400 two-moons points with Gaussian coordinate noise of standard deviation 0.08. We revealed one uniformly sampled label per class and scored predictions on the

other 398 points. Euclidean 1-NN and our NumPy LGC implementation used the same coordinates and revealed labels in each trial.

At our fixed setting of $\sigma = 0.15$ and $\alpha = 0.99$, mean error across seeds was 0.45% for LGC (sample SD 0.54 percentage points), compared with 22.36% for 1-NN (SD 4.22 percentage points). LGC had lower error in every paired trial. Its per-seed accuracies also matched the independent fixed-point check described in Section 3.

A descriptive sweep held α, data, and revealed labels fixed. LGC mean error was 0.15%, 0.10%, 0.45%, 14.07%, and 32.59% at $\sigma = 0.05, 0.10, 0.15, 0.30$, and 0.60, respectively. The results show that label propagation can exploit the curved structure, and that the advantage depends on the graph bandwidth. The sweep describes sensitivity on these trials. Our results concern generated toy data with a label from each class, so their scope is narrower than the paper's image and text studies.

4 Critical discussion

[D to write: assumptions of the graph construction, model-selection gap, sensitivity to σ, scalability of the dense solve. C's `docs/c_paper_evidence.md` *limitations list is the review-approved raw material. Target ≈0.9 page.]* ***[TODO: placeholder]***

5 Conclusion

[To be drafted jointly 25–27 Nov.] ***[TODO: placeholder]***

References

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